AMA572 Lecture 4.

Leland-Toft Endogenous Default Model

1 Basic Setting of the Model

1.1 Asset Model

Leland and Toft (1996) assume that under a risk-adjusted probability P, the asset value of the firm V_t follows

$$\frac{dV_t}{V_t} = (r - \delta)dt + \sigma dW_t,$$

whose solution is given by

$$V_t = V_0 \exp\left\{(r - \delta - \frac{1}{2}\sigma^2)t + \sigma W_t\right\},$$

with r being the constant risk-free interest rate, δ the total payout rate to the firm's investors (including both bond and equity holders). Note that under the above risk-adjusted probability V_t is a martingale after proper discounting: $V_t = E\left[e^{-(r-\delta)(T-t)}V_T|\mathcal{F}_t\right]$, where $\mathcal{F}_t$ is the information up to time t.

1.2 Debt Issuing and Coupon Payments

The setting of debt issuing and coupon payment follows Leland (1994) and Leland and Toft (1996). In the time interval $(t, t + dt)$, the firm issues new debt with par value pdt, and maturity profile φ, where $\varphi(v) = me^{-mv}$, i.e. the maturity of a specific bond is chosen randomly according to an exponentially distributed random variable with mean $1/m$. At any time t, the par value of debt maturing in the next moment $(s, s + ds)$ is given by

$$\left(\int_{-\infty}^{t} p\varphi(s-u)du\right)ds = \left(\int_{-\infty}^{t} pme^{-m(s-u)}du\right)ds = pe^{-m(s-t)}ds. \quad (1)$$

The above equation (1) has two implications. First, letting $s = t$ in (1) we have at time t the par value of the debt maturing at the next moment $(t, t + dt)$ is given by pdt, which is the same as the par value of the newly-issued debt, achieving a static balance. Secondly, using (1), at time t the par value of all pending debt (with maturity from t to ∞) is a constant, equal to

$$P = \int_{t}^{+\infty} pe^{-m(s-t)}ds = p/m.$$

We also assume that before maturity bondholders will receive coupons at rate ρ until default.

At each moment the firm faces two cash outflows and two cash inflows. The two cash outflows are after-tax coupon payment $(1-\kappa)\rho P dt$ and due principal pdt, where κ is the tax rate; the two cash inflows are $b_t dt$ from selling new debts, where b_t is the price of the total newly issued bonds, and the total payout from the asset $\delta V_t dt$. If the total cash inflow $(\delta V_t + b_t)dt$ is greater than the cash outflow $((1-\kappa)\rho P+p)dt$, we assume that the difference of the two goes to the equity holders as dividends; otherwise, additional equity will be issued to fulfill the due liability. Note that the difference $[((1-\kappa)\rho P + p) - (\delta V_t + b_t)]dt$ is an infinitesimal quantity. Thus, such a financing strategy is feasible as long as the total equity value remains positive before bankruptcy. Therefore, we need to impose the limited liability constraint on the dynamic of the firm's equity.

1.3 Default Payments

As in first passage models, we assume that the default occurs at time $\tau = \inf\{t \geq 0 : V_t \leq V_B\}$. Upon default, the firm loses $(1-\alpha)$ of V_τ, due to reorganization of the firm, and the debtholders as a whole get the rest of the value left, αV_τ, after reorganization.

To find the total debt value, total equity value, and optimal leverage ratio, it is not necessary to spell out the recovery value for individual bonds with different maturities. However, to model credit spreads we need to specify how the remaining asset of the firm αV_τ is distributed among bonds with different maturities. Three standard assumptions are recovery at a fraction of par value, of market values, and of corresponding treasury bonds. Here we shall use the assumption of recovery at a fractional treasury bonds. We use this assumption because it makes analytical calculation easier. More precisely, upon default the payoff of a bond with unit face value and maturity $T > \tau$ is given by

$$ce^{-r(T-\tau)}, \;\; 0 \leq c \leq 1.$$

To determine c, we shall match the total payment to bondholder with the total remaining asset αV_τ. By the memoryless property of exponential distribution, the conditional distribution of a bond's maturity does not change given its maturity is beyond τ. Thus, we have by (1)

$$\int_\tau^{+\infty} ce^{-r(s-\tau)} \cdot pe^{-m(s-\tau)} ds = \alpha V_\tau,$$

yielding

$$c = \frac{m+r}{m}\frac{\alpha V_\tau}{P} = \frac{m+r}{m}\frac{\alpha V_B}{P}.$$

To make sure that $0 \leq c \leq 1$, we shall impose that

$$\frac{m+r}{m}\frac{\alpha V_B}{P} \leq 1. \tag{2}$$

We will see that this assumption is satisfied by the optimal default barrier; see Theorem 1.

We should emphasize that this recovery assumption is irrelevant if one is only interested in understanding debt values as a whole, but is relevant when we consider credit spreads, which in turn depend on cash flows to bonds with different maturities. In particular, our results about optimal default boundary and optimal leverage level will not be affected by the recovery assumption.

1.4 Debt, Equity, and Market Value of the Firm

At time 0 the price of a bond with face value 1 and maturity T when the firm asset $V_0 = V$ is given by

$$\begin{aligned}
B(V;V_B,T) &= E\left[e^{-rT}\mathbf{1}_{\{\tau>T\}} + e^{-r\tau}\cdot ce^{-r(T-\tau)}\mathbf{1}_{\{\tau\leq T\}}\right] + E\left[\int_0^{\tau\wedge T}\rho e^{-rs}ds\right] \\
&= e^{-rT}P[\tau\geq T] + cE[e^{-rT}\mathbf{1}_{\{\tau\leq T\}}] + \frac{\rho}{r}(1-E[e^{-r(\tau\wedge T)}]) \\
&= \left(1-\frac{\rho}{r}\right)e^{-rT}P[\tau\geq T] + cE\left[e^{-rT}\mathbf{1}_{\{\tau\leq T\}}\right] + \frac{\rho}{r}\left(1-E\left[e^{-r\tau}\mathbf{1}_{\{\tau\leq T\}}\right]\right).
\end{aligned}$$

In our endogenous model, both P (which is related to the optimal debt/equity ratio) and the endogenous default barrier V_B will be decision variables.

According to the Modigliani-Miller theorem, the total market value of the firm is the asset value plus tax benefits and minus the bankruptcy costs. More precisely, the market value of the firm at time 0 is

$$\begin{aligned}
v(V;V_B) &= V + E\left[\int_0^{\tau}\kappa\rho P e^{-rt}dt\right] - (1-\alpha)E[V_\tau e^{-r\tau}] \\
&= V + \kappa\rho P\cdot E\left[\int_0^{\tau}e^{-rt}dt\right] - (1-\alpha)V_B E[e^{-r\tau}].
\end{aligned}$$

Note that we have to put a constraint on ρ,

$$\frac{r}{\kappa} > \rho; \tag{3}$$

otherwise, the total tax benefit, $E[\int_0^\tau \kappa\rho P e^{-rt}dt] \leq \kappa\rho P\int_0^\infty e^{-rt}dt = \frac{\kappa\rho}{r}P$, could exceed the face value of the debt P, which may create a tax abitrage

opportunity. The total equity value equals to the total value of the firm less the total value of bonds,

$$S(V;V_B) = v(V;V_B) - D(V;V_B),$$

where the total debt $D(V;V_B)$ is equal to

$$D(V;V_B) = P\int_0^{+\infty} me^{-mT}B(V;V_B,T)dT.$$

We should point out that in this static model (thanks to the exponential maturity profile), the total firm value, total debt and equity values are all Markovian, and are independent of the time horizon.

1.5 Preliminary Results for Debt and Equity Values

Similarly the total market value of the firm $v(V;V_B)$ can be computed similarly in a hwk problem, and the total equity of the firm is given by $S(V;V_B) = v(V;V_B) - D(V;V_B)$.

Proposition 1. The total market value of the firm is given by

$$v(V;V_B) = V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right](V_B/V)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi}. \quad (4)$$

The total debt value of the firm is given by

$$D(V;V_B) = P\frac{\rho+m}{r+m} - P\left(\frac{\rho}{r} + \left(1 - \frac{\rho}{r} - \frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right)(V_B/V)^{\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi}. \quad (5)$$

The total equity value of the firm is given by

$$\begin{aligned} S(V;V_B) &= v(V;V_B) - D(V;V_B) \\ &= V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right](V_B/V)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi} - P\frac{\rho+m}{r+m} \\ &\quad + P\left[\frac{\rho}{r} + \left(1 - \frac{\rho}{r} - \frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right](V_B/V)^{\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi}. \end{aligned} \quad (6)$$

Proof. The market value of the firm at time 0 is

$$\begin{aligned} v(V;V_B) &= V + \kappa\rho P \cdot E\left[\int_0^\tau e^{-rt}dt\right] - (1-\alpha)V_B E[e^{-r\tau}] \\ &= V + \kappa\rho P \cdot E\left[\frac{1}{r}(1-e^{-r\tau})\right] - (1-\alpha)V_B E[e^{-r\tau}] \\ &= V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\cdot E[e^{-r\tau}]. \end{aligned}$$

Since

$$E[e^{-r\tau}] = \exp\left\{\log(V_B/V)\cdot\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\right\},$$

where $\xi = \frac{r-\delta}{\sigma^2} - \frac{1}{2}$, we must have (4).

We know from the previous discussions that the Laplace transform of a bond price, $B(V;V_B,T)$, is given by

$$\begin{aligned}&\int_0^\infty e^{-\beta T}B(V;V_B,T)dT\\ &= \frac{\rho+\beta}{\beta(r+\beta)} - \left(\frac{\rho}{r}\frac{1}{\beta} + \left(1-\frac{\rho}{r}-c\right)\frac{1}{r+\beta}\right)\cdot\exp\left\{\log(V_B/V)\cdot\left(\sqrt{\xi^2+\frac{2}{\sigma^2}(\beta+r)}+\xi\right)\right\},\end{aligned}$$

where

$$\xi = \frac{r-\delta}{\sigma^2} - \frac{1}{2}, \qquad c = \frac{m+r}{m}\frac{\alpha V_B}{P}.$$

Therefore, the value of total debt at time 0 is

$$\begin{aligned}D(V;V_B) &= P\int_0^{+\infty} me^{-mT}B(V;V_B,T)dT\\ &= P\frac{\rho+m}{r+m} - P\left(\frac{\rho}{r} + \left(1-\frac{\rho}{r}-\frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right)\\ &\quad\times\exp\left\{\log(V_B/V)\cdot\left(\sqrt{\xi^2+\frac{2}{\sigma^2}(m+r)}+\xi\right)\right\}\\ &= P\frac{\rho+m}{r+m} - P\left(\frac{\rho}{r} + \left(1-\frac{\rho}{r}-\frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right)\cdot(V_B/V)^{\sqrt{\xi^2+2(m+r)/\sigma^2}+\xi},\end{aligned}$$

which is (5). The combination of equation (4) and (5) yields that the total equity value of the firm is given by (6).

Remark: Note that we are able to compute the total debt value, total equity value, and the total market value analytically, although we can only compute a given bond price via Laplace transforms. This is made possible thanks to our assumption of the exponential maturity profile.

2 Optimal Capital Structure and Endogenous Default

In this section, we consider the problem of optimal capital structure and optimal endogenous default barrier, i.e. the choices of optimal debt level P and bankrupt trigger V_B.

2.1 The Solution of a Two-Stage Optimization Problem

Deciding optimal P and choosing optimal V_B are two entangled problems, and cannot be separated easily. For example, when a firm chooses P to maximize the total firm value at time 0, the decision on P obviously depends on the bankruptcy trigger level V_B. Similarly, after the debt being issued, the equity holders will choose an optimal V_B, which of course depends on how much debt the firm has. Here we shall choose P and V_B according to a two-stage optimization procedure as in Leland (1994) and Leland and Toft (1996).

More precisely, for a fixed P, the equity holders find the optimal default barrier by maximizing the equity value, subject to the limited liability constraint. It is the equity value that is maximized, because after the debt being issued it is the equity holders who control the firm. Mathematically, the equity holders find the best V_B by solving

$$\max_{V_B} S(V; V_B) \tag{7}$$

subject to the limited liability constraint

$$S(V'; V_B) \geq 0, \text{ for all } V' \geq V_B \geq 0. \tag{8}$$

The limited liability constraint is imposed so that the equity value is always nonnegative for any future firm asset value V', as long as the asset value V' is above the bankruptcy level V_B. The is the first stage optimization problem.

Clearly the optimal $V_B^* \equiv V_B^*(P)$ to be chosen by the equity holders will depend on P. At time 0, the firm will conduct the second stage optimization to maximize the firm value, in anticipation of what the equity holders will do later. More precisely, the second stage optimization is that

$$\max_{P} v(V; V_B^*(P)). \tag{9}$$

The two-stage optimization problem partly arises due to the conflict of interests between debt and equity holders. It is obvious that choosing the leverage level P and bankrupt trigger V_B simultaneously will lead to a better market value of the firm. Leland (1998) uses the difference of the two to explain agency costs due to conflicts between the equity and debt holders. Obviously the two-stage optimization is only a rough approximation to complicated decision and negotiation between debt and equity holders. Here is the result for the two-stage optimization.

Theorem 1. (a) The solution of

$$\frac{\partial S(V; V_B^*)}{\partial V}|_{V=V_B^*} = 0 \tag{10}$$

is given by $V_B^* = \epsilon P$, where

$$\epsilon = \frac{\frac{\rho+m}{r+m}\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right) - \frac{\kappa\rho}{r}\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)}{1+(1-\alpha)\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)+\alpha\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)}.$$

(b)

$$0 < \epsilon < \frac{1}{\alpha}\frac{\rho+m}{r+m}, \tag{11}$$

and, therefore, the default barrier V_B^* satisfies (2).

(c) The first stage optimization: Given the debt level P, the optimal barrier level solving (7) with the constraint in (8) is given by $V_B^* = \epsilon P$.

(d) The second stage optimization: After plugging the optimal $V_B^* = \epsilon P$ into the second stage optimization, we have that $v(V; \epsilon P)$ is a concave function of $P \in (0, V/\epsilon)$, which implies that we can find a unique optimal debt level P for the problem (9).

Proof. (a) It follows from equation (6) that

$$\begin{aligned}
&\frac{\partial S(V;V_B)}{\partial V} \\
&= 1 - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\left(\frac{V_B}{V}\right)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi-1}\left(-\frac{V_B}{V^2}\right) \\
&\quad + P\left[\frac{\rho}{r} + \left(1-\frac{\rho}{r}-\frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right]\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)\left(\frac{V_B}{V}\right)^{\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi-1}\left(-\frac{V_B}{V^2}\right).
\end{aligned}$$

Hence,

$$\begin{aligned}
&\frac{\partial S(V;V_B)}{\partial V}\Big|_{V=V_B} \\
&= 1 - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\left(-\frac{1}{V_B}\right) \\
&\quad + P\left[\frac{\rho}{r} + \left(1-\frac{\rho}{r}-\frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right]\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)\left(-\frac{1}{V_B}\right).
\end{aligned}$$

Letting

$$\frac{\partial S(V;V_B)}{\partial V}\Big|_{V=V_B} = 0,$$

we obtain

$$1+\frac{\kappa\rho}{r}P\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\frac{1}{V_B}+(1-\alpha)\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)$$
$$-P\left[\frac{\rho}{r}+\left(1-\frac{\rho}{r}\right)\frac{m}{r+m}\right]\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)\frac{1}{V_B}+\alpha\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)=0,$$

which yields

$$V_B^*=\frac{\left[\frac{\rho}{r}+\left(1-\frac{\rho}{r}\right)\frac{m}{r+m}\right]\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)-\frac{\kappa\rho}{r}\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)}{1+(1-\alpha)\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)+\alpha\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)}P.$$

So $V_B^*=\epsilon P$.

(b) See a homework problem.

(c) This part is the most difficult part; see the paper by Chen and Kou (2009).

(d) By equation (4) we have

$$\begin{aligned}v(V;\epsilon P)&=V+\frac{\kappa\rho}{r}P-\left[\frac{\kappa\rho}{r}P+(1-\alpha)\epsilon P\right](\epsilon P/V)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi}\\&=V+\frac{\kappa\rho}{r}P-\left[\frac{\kappa\rho}{r}+(1-\alpha)\epsilon\right](\epsilon/V)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi}\cdot P^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi+1}.\end{aligned}$$

Since

$$\sqrt{\xi^2+\frac{2r}{\sigma^2}}+\xi+1>\sqrt{\xi^2}+\xi+1\geq 1,$$

it follows that

$$P^{\left(\sqrt{\xi^2+\frac{2r}{\sigma^2}}+\xi+1\right)}\text{ is convex,}$$

and consequently

$$-\left[\frac{\kappa\rho}{r}+(1-\alpha)\epsilon\right]\left(\frac{\epsilon}{V}\right)^{\sqrt{\xi^2+\frac{2r}{\sigma^2}}}\cdot P^{\sqrt{\xi^2+\frac{2r}{\sigma^2}+\xi+1}}.$$

The expression above is concave.

Since $V+\frac{\kappa\rho}{r}P$ is linear and the sum of a linear function and a concave function is still concave, it follows that the function $v(V;\epsilon P)$ is concave.

2.2 The Optimal Capital Structure

Table 2 shows the effect of various parameters on the optimal leverage level. The parameters are $\kappa = 35\%$, $r = 8\%$, $\rho = 8.162\%$, $\delta = 6\%$, $V_0 = 100$. The riskless rate $r = 8\%$ is close to the average historical short-term treasury rate during 1973-1998, and the coupon rate $\rho = 8.162\%$ is the par coupon rate for risk-free bonds with semi-annual coupon payments when the continuously compounded interest rate is 8%.

Consistent with our intuition, the table shows that the optimal leverage ratio is an increasing function of the fractional remaining asset α and average maturity profile $1/m$, and is a decreasing function of diffusion volatility σ.

	$m^{-1} = 0.5$		$m^{-1} = 1$		$m^{-1} = 2$		$m^{-1} = 5$	
	$\sigma = 0.2$	$\sigma = 0.4$	$\sigma = 0.2$	$\sigma = 0.4$	$\sigma = 0.2$	$\sigma = 0.4$	$\sigma = 0.2$	$\sigma = 0.4$
$\alpha = 5\%$	7.14%	1.11%	11.19%	2.37%	17.56%	5.04%	30.67%	12.91%
$\alpha = 25\%$	13.78%	3.21%	18.31%	5.28%	25.08%	9.13%	38.41%	19.07%
$\alpha = 50\%$	25.44%	9.22%	30.30%	12.63%	37.29%	18.27%	50.52%	31.19%

However, the Leland-Toft model cannot explain why many high-tech companies have no debt at all.

HWK

1. Prove (11).